

Matthew Mercado

Full-Stack Software Engineer · AI Applications & Developer Tools

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Washington, DC Metro Area · Active TS/SCI with Full-Scope Polygraph · Fluent in English and Spanish

Full-stack software engineer with 9 years of professional experience, known for polished, accessible interfaces and thoughtful UX. Currently building LLM agent platforms and AI developer tools in TypeScript and Python; founder of two SaaS products.

EXPERIENCE

National Security Agency · U.S. Department of Defense

Aug 2024 – Present

AI Agents Software Engineer

Apr 2026 – Present

- Built 3 of the 4 production Model Context Protocol (MCP) servers for an agency-wide LLM platform in Python/FastMCP, handling thousands of tool calls per month through an enterprise Svelte chat interface.
- Owned code review for the platform's 15 agent-tooling repositories (20+ merge requests per week).
- Shipped an air-gapped Electron/TypeScript IDE for AI coding agents on Linux and Windows with offline mirrors, a model gateway, PKI authentication, and in-app updates; used by dozens of employees before it was advertised.
- Wrote agent skills, workflows, and an agentic-engineering onboarding course that enable AI-assisted development across AI teams, including for developers new to software engineering.
- Built a multi-step research agent and knowledge-repository MCP server that searches multiple sources and returns one cited answer, replacing step-by-step manual lookups.
- Enabled plain-English chart and diagram generation in seconds through schema-validated visualization tools exposed as MCP servers.

Technical Lead

Jan 2025 – Apr 2026

- Architected a self-service ingestion platform (Next.js, TypeScript, FastAPI, PostgreSQL, Redis) for 100,000+ entity datasets that previously required bespoke scripts; embedded it in 3 partner tools through an API-backed widget and earned a division performance award.
- Cut 10,000-entity processing from 7 minutes to under 1 minute with batched writes and optimized aggregation.
- Made a nightly purge job 7x faster by scanning only files due for deletion instead of every stored file.
- Sped React rendering up to 3x on large-list views by moving state to Zustand and adding list virtualization.
- Cut build time 6x and new-feature bug reports ~80% in the following month by introducing CI/CD with Vitest and pytest suites and an Argo CD GitOps pipeline deploying Docker containers to Kubernetes.
- Mentored 5 engineers across 3 teams through onboarding documentation, architecture reviews, and pairing.

Software Engineer

Aug 2024 – Jan 2025

- Cut analyst task-completion time 90% by building a React/TypeScript, Java/Spring Boot, and MariaDB workflow application to replace a legacy system.
- Delivered 3 months ahead of a one-year plan and earned a division team award; four partner agencies requested the application and one began implementing it.
- Led an 11-person government and contractor team to 100% on-time milestones as acting product lead, owning backlog, requirements, and release timelines.
- Raised Section 508 accessibility scores 78% (to about 4.8/5) with accessible replacements for legacy UI.

eMediCall

Nov 2024 – Present

Contract Software Engineer (part-time)

- Delivered a React Native (Expo) mobile frontend for a medical messaging application on existing PHP endpoints, with Google sign-in, camera workflows, and push notifications.

Escape This Frederick · Frederick, MD

Jul 2017 – Aug 2024

Software Engineer & Manager

- Developed the company's digital waiver application (Next.js, React, Tailwind CSS, Prisma, PostgreSQL) with Bookeo and Mailchimp integrations, capturing every guest for remarketing.
- Doubled website conversion and reduced bounce rate 35% after redesigning escapethisfrederick.com.
- Engineered Arduino/C++ and PLC-controlled electronic puzzles that eliminated manual puzzle resets.

Box · Remote

May 2022 – Aug 2022

Software Engineer Intern, Preview Team

- Built PHP social-link unfurling across all file types, generating rich previews for thousands of files shared daily.
- Upgraded analytics menu components with React, Redux, and Sass.
- Reduced manual QA by extending a behavior-driven test harness with Gherkin, Cucumber, and WebdriverIO.

INDEPENDENT PRODUCTS

Escape Director · escapedirector.com

May 2023 – Present

Founder & Lead Software Engineer

- Built and operate an escape-room management SaaS (TanStack Start, React, TypeScript, Express, Prisma, PostgreSQL, AWS S3, Stripe) that has recorded 7,600+ games with 99.95% uptime over the trailing 12 months.
- Shipped offline room operation with an IndexedDB store, verified offline access grants, and a durable, idempotent outbox, so rooms keep running through internet outages and completed games sync once connectivity returns.
- Developed an LLM analytics assistant on OpenAI models through Vercel AI Gateway that answers business questions in natural language and customizes operators' dashboards through schema-validated tool calls.
- Built an LLM evaluation harness on the production tool flow that compared 4 models across 3 benchmark rounds (72 turns per model), using trace grading, blinded answer review, and spend limits to select the production model on quality, latency, and cost.
- Released an open-source (MIT) C++ Arduino device SDK connecting physical props to live games, with state, command, and completion APIs plus automatic pairing and reconnection.

Waiver Director · waiverdirector.com

Mar 2026 – Present

Founder & Lead Software Engineer

- Commercialized the Escape This waiver tool as a multi-tenant guest-CRM SaaS (SvelteKit, TypeScript, Convex) that captures every participant, not just the booker, with immutable audit trails and automated guest follow-up.
- Integrated Bookeo and Xola bookings, Mailchimp and Constant Contact sync, Clerk authentication, Stripe billing, and role-based team access.

EDUCATION & CERTIFICATIONS

Mount St. Mary's University · B.S. Computer Science, summa cum laude, May 2023 · Mariam Award (highest GPA in Computer Science) · Upsilon Pi Epsilon

Frederick Community College · A.S. STEM, Engineering Concentration, May 2019

MIT Lincoln Laboratory · iDiscovery: Agentic AI certificate, Apr 2026

SKILLS

AI & LLM: LLM agents, MCP/FastMCP, RAG, tool calling, LLM evaluation, Vercel AI SDK/Gateway, OpenAI API

Languages: TypeScript, JavaScript, Python, Java, SQL, C++, PHP

Frontend: React, Next.js, TanStack Start/Query, SvelteKit, Zustand, Redux, Tailwind CSS, Electron, React Native

Backend & Data: Node.js/Express, FastAPI, Spring Boot, REST APIs, Prisma, Convex, PostgreSQL, Redis

Cloud & DevOps: Docker, Kubernetes, Helm, Argo CD, GitOps, CI/CD, AWS (EKS, ECR, EC2, S3), Git, Linux

Testing: Vitest, pytest, Playwright, Cucumber, WebdriverIO